Project Title
Event Management System

Project Report submitted for the partial fulfilment of grade for the course

Data Structures and Algorithms -3 (25CS2103E)

in

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ABSTRACT

The Event Management System is a terminal-based Java application designed to efficiently manage, organize, search, and optimize college and organizational events. The system stores event details such as title, date, venue, category, capacity, and participants, allowing users to add, search, update, and schedule events. Advanced algorithms from all six course outcomes are integrated into the system. CO1 is addressed through algorithm selection and complexity analysis for event-related problems. CO2 uses KMP, Z-function, Rabin-Karp, and suffix-based searching for event and category matching. CO3 applies dynamic programming, including bitmask DP, for optimizing event schedules and venue allocation. CO4 uses network-flow techniques such as Ford-Fulkerson or Edmonds-Karp for capacity-constrained venue and participant assignment. CO5 models scheduling as an NP-hard optimization problem and applies approximation techniques to obtain practical solutions. CO6 incorporates randomized algorithms and parallel-processing concepts for scalable searching, sorting, sampling, and performance evaluation. The system runs through a command-line interface using Java.

Signature of the Guide